🚀 Complete Deployment Guide — ZoneAnalyzer

📁 Step 1: Upload Files (Local PowerShell)

```powershell
# Create folders on server
ssh hds@100.127.9.127 "mkdir -p ~/Webapp/ZoneAnalyzer2 ~/Webapp/backend/models"
```

```powershell
# Upload backend files
scp C:\code\git\Senior_project\Webapp\backend\main.py C:\code\git\Senior_project\Web
```

```powershell
# Upload models folder & database
scp -r C:\code\git\Senior_project\Webapp\backend\models hds@100.127.9.127:~/Webapp/b
scp C:\code\git\Senior_project\Webapp\backend\senior_project.db hds@100.127.9.127:~/
```

```powershell
# Upload frontend files
scp -r C:\code\git\Senior_project\Webapp\ZoneAnalyzer2\src hds@100.127.9.127:~/Webap
scp C:\code\git\Senior_project\Webapp\ZoneAnalyzer2\package.json C:\code\git\Senior_
```

🖥️ Step 2: SSH into Server

```powershell
ssh hds@100.127.9.127
```

🔄 Step 3: Setup Backend (First time only)

```bash
```

```bash
cd ~/Webapp/backend
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
pip install --upgrade ultralytics
```

⚙️ Step 4: Set Frontend API URL & Build

```bash
# Set API URL to domain
echo "VITE_API_BASE_URL=https://paniti-jupyter.cra.ac.th/zoneanalyzer/api" > ~/Webap

# Set image URL to domain
sed -i 's|/uploaded_images/|https://paniti-jupyter.cra.ac.th/zoneanalyzer/uploaded_i
sed -i 's|/uploaded_images/|https://paniti-jupyter.cra.ac.th/zoneanalyzer/uploaded_i

# (base: '/zoneanalyzer/' already set in vite.config.ts)

cd ~/Webapp/ZoneAnalyzer2
npm install
npm run build
```

🚀 Step 5: Start Both Services

```bash
```

```
# Kill old processes
pkill -f uvicorn
pkill -f "npx serve"
sleep 1

# Start backend on port 6014
cd ~/Webapp/backend && nohup .venv/bin/uvicorn main:app --host 0.0.0.0 --port 6014 >

# Start frontend on port 6015
nohup npx serve -s ~/Webapp/ZoneAnalyzer2/build -l tcp://0.0.0.0:6015 > ~/Webapp/fro

# Verify both running
sleep 3
curl -s -o /dev/null -w "Backend: %{http_code}\n" http://localhost:6014
curl -s -o /dev/null -w "Frontend: %{http_code}\n" http://localhost:6015
```

🌐 Step 6: Open Browser

```
https://paniti-jupyter.cra.ac.th/zoneanalyzer/
```

> ไม่ต้องทำ SSH tunnel แล้ว! ใครก็เปิดได้เลย

🔄 Re-deploy After Code Changes

powershell

```
# 1. Upload changed files (local PowerShell)
scp C:\code\git\Senior_project\Webapp\backend\main.py hds@100.127.9.127:~/Webapp/bac
```

bash

```
# 2. Restart backend (server) - only if backend changed
pkill -f uvicorn
cd ~/Webapp/backend && nohup .venv/bin/uvicorn main:app --host 0.0.0.0 --port 6014 >
```

bash

```bash
# 3. Rebuild & restart frontend (server) - only if frontend changed
cd ~/Webapp/ZoneAnalyzer2 && npm run build
pkill -f "npx serve"
nohup npx serve -s ~/Webapp/ZoneAnalyzer2/build -l tcp://0.0.0.0:6015 > ~/Webapp/fro
```

🛠️ Useful Debug Commands

```bash
# Check logs
tail -50 ~/Webapp/backend.log
tail -50 ~/Webapp/frontend.log

# Check running ports
ss -tlnp | grep 601

# Check running processes
ps aux | grep -E "uvicorn|serve"

# Check both services up
curl -s -o /dev/null -w "Backend: %{http_code}\n" http://localhost:6014
curl -s -o /dev/null -w "Frontend: %{http_code}\n" http://localhost:6015
```

📝 nginx Config (อาจารย์ต้องตั้งค่า — ทำแล้ว ✅)

```nginx
```

```
location /zoneanalyzer/ {
    proxy_pass http://localhost:6015/;
}

location /zoneanalyzer/api/ {
    proxy_pass http://localhost:6014/;
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto $scheme;
}

location /zoneanalyzer/uploaded_images/ {
    proxy_pass http://localhost:6014/uploaded_images/;
    proxy_set_header Host $host;
}
```